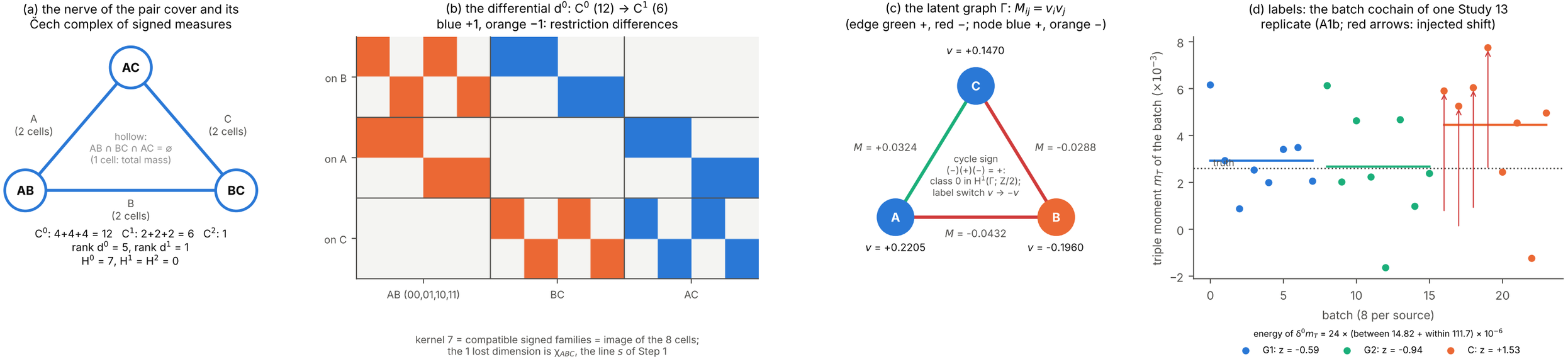
(a) the nerve of the pair cover and its Čech complex of signed measures
AC
A (2 cells)
C (2 cells)
hollow: AB ∩ BC ∩ AC = ∅ (1 cell: total mass)
AB
BC
B (2 cells)
C⁰: 4+4+4 = 12 C¹: 2+2+2 = 6 C²: 1
rank d⁰ = 5, rank d¹ = 1
H⁰ = 7, H¹ = H² = 0
(b) the differential d⁰: C⁰ (12) → C¹ (6)
blue +1, orange −1: restriction differences
on B
on A
on C
AB (00,01,10,11)
BC
AC
kernel 7 = compatible signed families = image of the 8 cells;
the 1 lost dimension is χ_ABC, the line s of Step 1
(c) the latent graph Γ: M_ij = v_i v_j
(edge green +, red −; node blue +, orange −)
v = +0.1470
C
M = +0.0324
M = -0.0288
cycle sign (−)(+)(−) = +: class 0 in H¹(Γ; Z/2); label switch v → −v
A
B
M = -0.0432
v = +0.2205
v = -0.1960
(d) labels: the batch cochain of one Study 13 replicate (A1b; red arrows: injected shift)
triple moment m_T of the batch (×10⁻³)
truth
batch (8 per source)
energy of δ⁰m_T = 24 × (between 14.82 + within 111.7) × 10⁻⁶
G1: z = -0.59
G2: z = -0.94
C: z = +1.53